

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

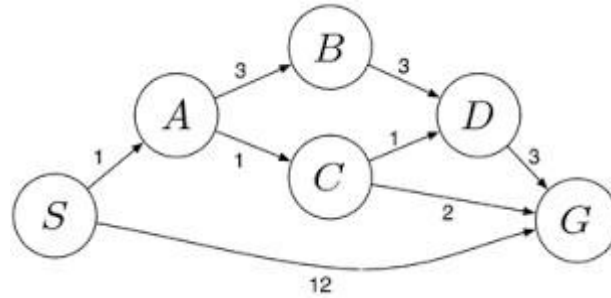
QUESTIONS: 5

Total Marks: 50

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an

associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I K.Pugazhendi (192311026) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

Answers for above Questions:

1. Given: 2 jugs, a 4-gallon one and 3gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water.

To Find: To get exactly 2 gallons of water into 4gallon jug

Solution:

Problem Formulation

State: (x, y) where x = amount of water in the 4gallon jug (0–4),
y = amount of water in the 3gallon jug (0–3)

Initial state: (0, 0)

Goal state: (2, y) for any y

Operators:

Fill the 4gallon jug from the pump $\rightarrow (4, y)$

Fill the 3gallon jug from the pump $\rightarrow (x, 3)$

Empty the 4gallon jug on the ground $\rightarrow (0, y)$

Empty the 3gallon jug on the ground $\rightarrow (x, 0)$

Pour water from the 3gallon jug into the 4gallon jug until the 4gallon jug is full or the 3gallon jug is empty

Pour water from the 4gallon jug into the 3gallon jug until the 3gallon jug is full or the 4gallon jug is empty

Solution Path

step	action	4 gallon	3 gallon
0	start	0	0
1	Fill 3 gallon	0	3
2	Pour 3 gallon into 4 gallon	3	0
3	Fill 3 gallon again	3	3
4	Pour from 3 into 4 gallon	4	2
5	Empty 4 gallon	0	2
6	Pour 2 gallon from 3 gallon into 4 gallon	2	0

Result: The 4gallon jug now holds exactly 2 gallons. Goal reached.

2. i) The Mars Rover receives information through its sensors:

- Camera images
- Rock and soil properties
- Temperature
- Atmospheric pressure
- Terrain obstacles
- GPS/Position data
- Wheel movement
- Battery level

(ii) Characterize the operating environment

Property	Description
Observability	Partially observable
Agents	Single agent
Deterministic	Mostly stochastic (uncertain terrain)
Sequential	Yes
Dynamic	Yes
Continuous	Yes

(iii) Actions

The rover can:

- Move forward/backward
- Turn left/right
- Capture images
- Collect soil samples
- Drill rocks
- Analyze samples
- Send data to Earth
- Recharge using solar panels
- Avoid obstacles

(iv) Performance Measure

Performance is evaluated by:

- Number of successful samples collected
- Accuracy of scientific observations
- Distance explored
- Energy efficiency
- Safe navigation
- Mission completion time

(v) Suitable Agent Architecture

Goal based + Utility based Agent

Reason

- Has mission goals.
- Chooses safest and most efficient route.
- Maximizes scientific value.
- Minimizes power consumption.
- Handles uncertain environments.

3. The objective of the 8 Queens problem is to place eight queens on an 8×8 chessboard in such a way that no queen attacks another queen. Since a queen can move horizontally, vertically, and diagonally, no two queens should be in the same row, column, or diagonal.

Problem Formulation

Initial State: The chessboard is empty, and no queens are placed.

State Space:

- A state represents the current arrangement of queens on the board.
- New states are generated by placing one queen at a time in a valid position.

Successor Function: From the current state, place a queen in the next column while ensuring it does not conflict with the already placed queens.

Goal Test: The goal is achieved when all 8 queens are successfully placed on the board without attacking each other.

Path Cost: Each queen placement has the same cost, so every solution has an equal path cost.

Search Strategy

The most suitable search strategy for this problem is Backtracking Search. It places queens one by one and checks whether the current placement is safe. If a conflict occurs, it goes back (backtracks) and tries another position. This method is efficient because it avoids exploring invalid arrangements.

4. When a customer books a cab using the OLA application, the system acts as a Goal-Based Agent. Its main objective is to help the customer reach the selected destination by finding the most suitable cab based on factors like availability, fare, distance, and customer preference.

A goal based agent makes decisions by considering the final goal. In this case, the goal is to book a suitable cab and complete the ride successfully. The system checks available cab options, allows the customer to choose one, confirms the booking, and guides the ride until the destination is reached.

Pseudocode

```
BEGIN
Enter source location
Enter destination
Display available cab types
(Mini, Micro, Sedan, Prime, Shared)
Customer selects a cab
IF the selected cab is available THEN
    Calculate the fare
    Confirm the booking
    Assign the nearest driver
    Display driver details
ELSE
    Show "Cab not available"
    Suggest another cab type
ENDIF
Track the ride until the destination
Generate the final bill
Complete the payment
END
```

5. Uniform Cost Search (UCS) is a search algorithm that always expands the node with the lowest accumulated path cost. It guarantees the optimal solution when all edge costs are positive.

Given Graph

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$

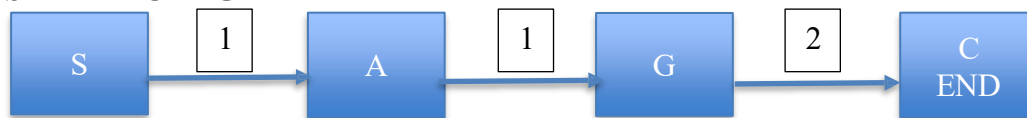
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Working of UCS

1. Start from node S with a cost of 0.
2. Expand S and add A (1) and G (12) to the priority queue.
3. Expand A (1) because it has the lowest cost. Add B (4) and C (2).
4. Expand C (2). From C, add D (3) and G (4).
5. Expand D (3). The path to G through D costs 6, which is higher than the existing cost 4, so it is ignored.
6. Expand B (4). It leads to D with a cost of 7, which is also ignored because a cheaper path already exists.
7. Finally, expand G (4). Since G is the goal node, the search stops.

Optimal Path

$S \rightarrow A \rightarrow C \rightarrow G$



Totalcost: $1 + 1 + 2 = 4$

Conclusion:

Using Uniform Cost Search, the least-cost path from the starting warehouse S to the destination warehouse G is:

$S \rightarrow A \rightarrow C \rightarrow G$

Minimum Cost = 4